BLOCKCHAIN LABORATORY

Experiment No. 4

Creation of Account in MetaMask and Transfer of Cryptocurrency

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1. Aim

To create accounts in MetaMask and transfer cryptocurrency from one account to another using the Sepolia Ethereum test network.

2. Theory

2.1. 1. MetaMask

MetaMask is a cryptocurrency wallet used to manage blockchain accounts and interact with decentralized applications. It provides an interface for viewing account balances, receiving and sending cryptocurrency, and signing blockchain transactions.

2.2. 2. Blockchain Account

A blockchain account is represented by a unique public address. The address can be shared with others to receive cryptocurrency. Transactions are authorized using the account's private cryptographic credentials.

2.3. 3. Sepolia Test Network

Sepolia is an Ethereum test network used for development and experimentation. The cryptocurrency used on the test network is test ETH, which does not represent real monetary value. Using a test network allows transactions to be performed without spending real cryptocurrency.

2.4. 4. Cryptocurrency Transaction

A cryptocurrency transaction transfers a specified amount of cryptocurrency from a sender address to a receiver address. The transaction is digitally signed by the sender and submitted to the blockchain network for confirmation.

2.5. 5. Gas Fee

Ethereum transactions require a network fee called *gas*. The fee compensates the network for processing the transaction. The sender must have sufficient ETH to cover both the amount being transferred and the applicable transaction fee.

2.6. 6. Block Explorer

A block explorer provides a public interface for viewing blockchain data. Transaction details such as the transaction hash, status, block number, sender, receiver, transferred value, and transaction fee can be verified using a Sepolia block explorer such as Etherscan.

3. Tools Used

- MetaMask browser wallet
- Sepolia Ethereum Test Network
- Sepolia PoW Faucet
- Sepolia Etherscan block explorer
- Web browser

4. Tasks Performed

4.1. Task 1 – Open MetaMask and Create the First Account

MetaMask was opened in the browser and the first account was created. The account was named `Nikhil`. The wallet was configured to use the Sepolia test network.

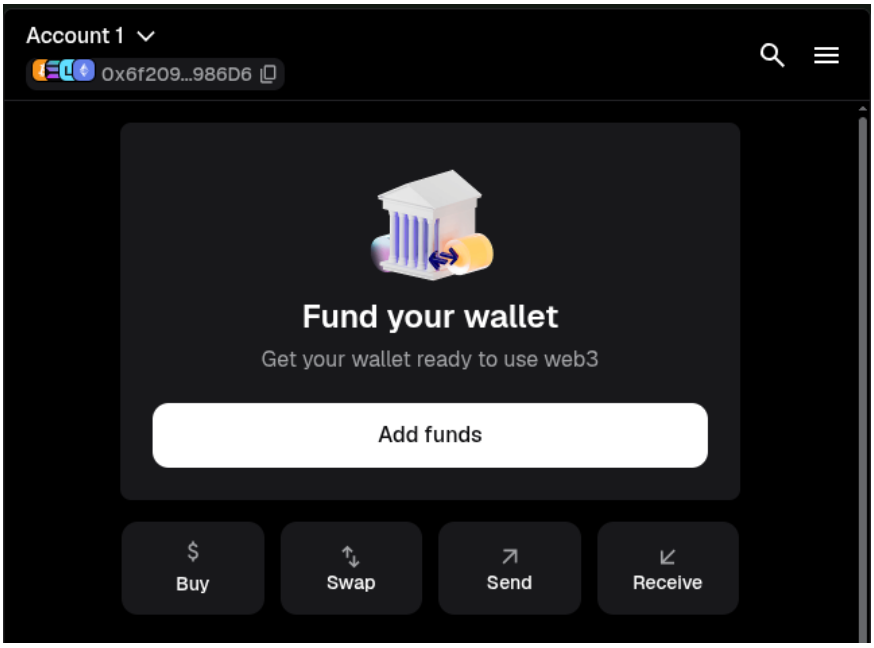


Fig. 1 – MetaMask Account 1 before receiving test ETH

The wallet initially showed no Sepolia ETH, so test cryptocurrency was obtained from a Sepolia faucet before performing the transfer.

4.2. Task 2 – Create a Second Account

A second MetaMask account was created and named `Sunchu`. The Accounts section shows both accounts with their respective wallet addresses.

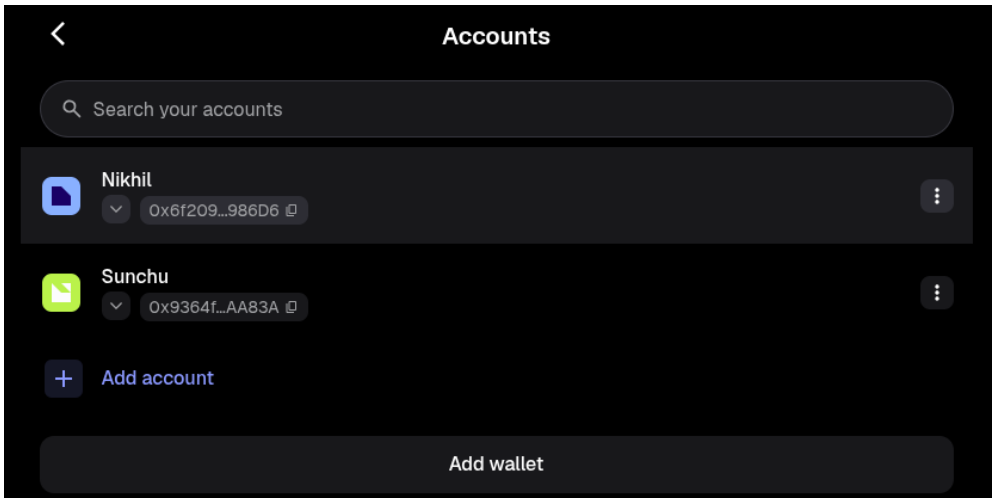


Fig. 2 – Two MetaMask accounts: Nikhil and Sunchu

The two accounts used in the experiment were:

Account	Purpose
Nikhil	Sender account
Sunchu	Receiver account

4.3. Task 3 – Verify the Initial Sender Balance

Before receiving test cryptocurrency, the Nikhil account showed a zero SepoliaETH balance on the Sepolia network.

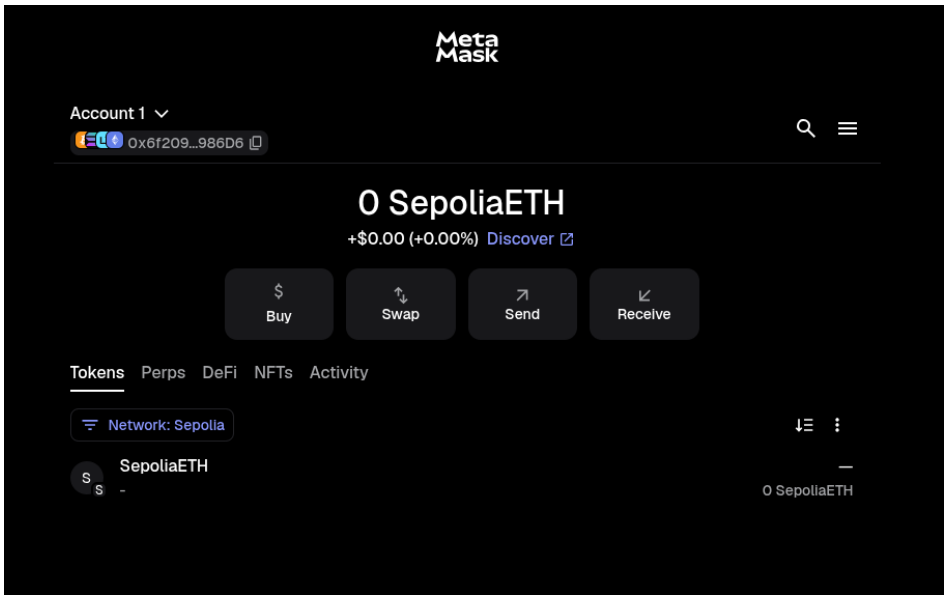


Fig. 3 – Initial SepoliaETH balance of Account 1

4.4. Task 4 – Obtain Test ETH from Sepolia Faucet

Test ETH was requested for the sender account using the Sepolia PoW Faucet. The faucet confirmed a reward of 2.5 SepETH for the wallet address corresponding to the Nikhil account.

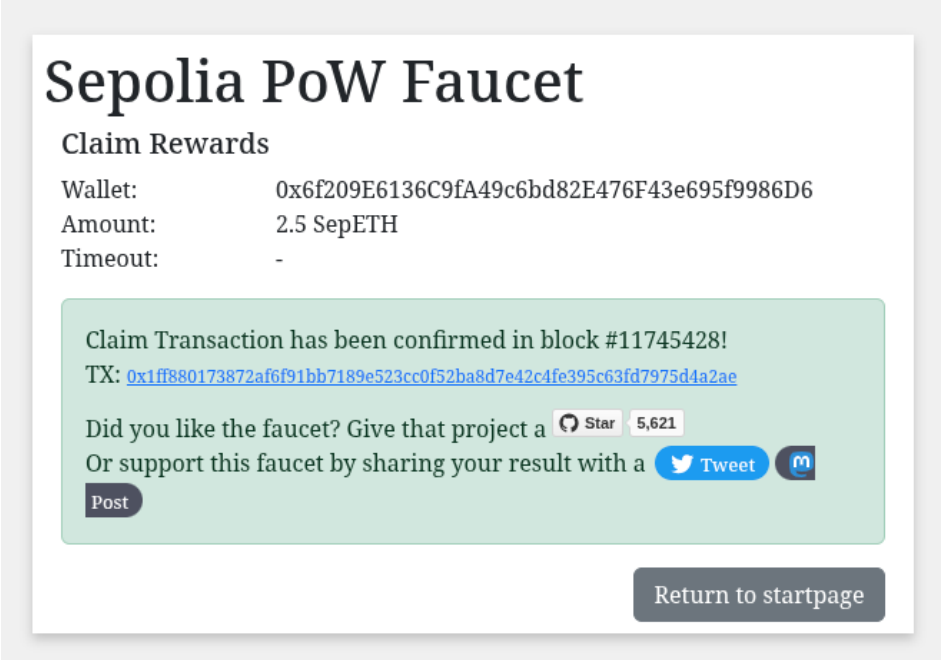


Fig. 4 – Sepolia PoW Faucet confirming the test ETH claim

The faucet displayed that the claim transaction was confirmed in block `#11745428`. After the claim, the Nikhil account displayed 2.5 SepoliaETH.

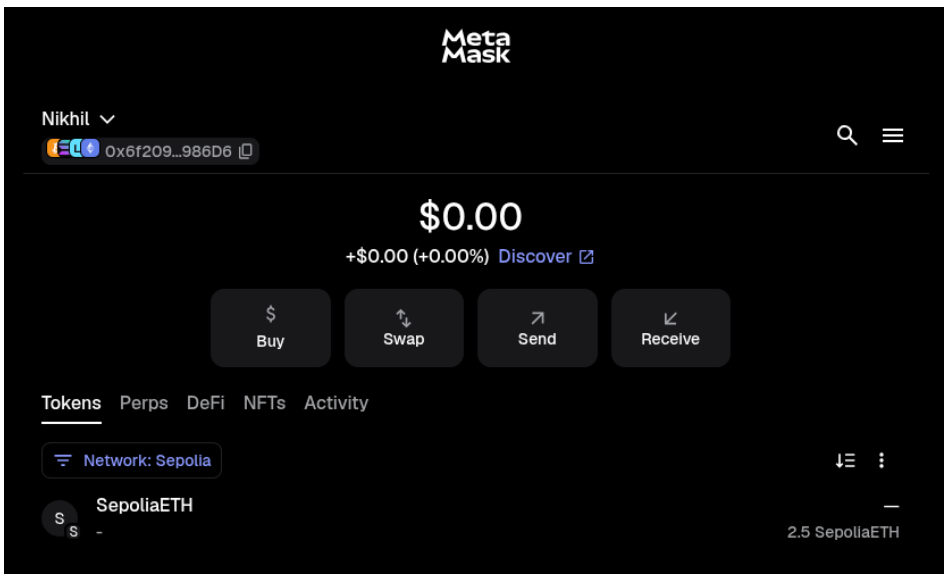


Fig. 5 – Nikhil account showing 2.5 SepoliaETH received from the faucet

4.5. Task 5 – Transfer Cryptocurrency from Account 1 to Account 2

After obtaining test ETH, the Nikhil account was used as the sender and the Sunchu account was used as the receiver. An amount of **0.01 SepoliaETH** was transferred.

The transaction was confirmed successfully on the Sepolia network.

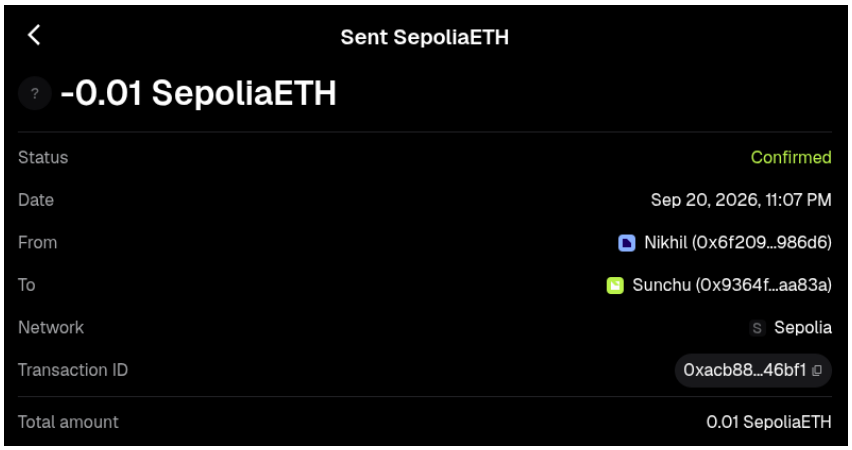


Fig. 6 – MetaMask transaction showing 0.01 SepoliaETH sent from Nikhil to Sunchu

The transaction details shown in MetaMask include:

- **Status:** Confirmed
- **Network:** Sepolia
- **From:** Nikhil
- **To:** Sunchu
- **Total amount:** 0.01 SepoliaETH

4.6. Task 6 – Verify the Receiver Account

The Sunchu account was opened after the transfer. The wallet displayed a balance of **0.0100 SepoliaETH**, confirming that the transferred cryptocurrency was received successfully.

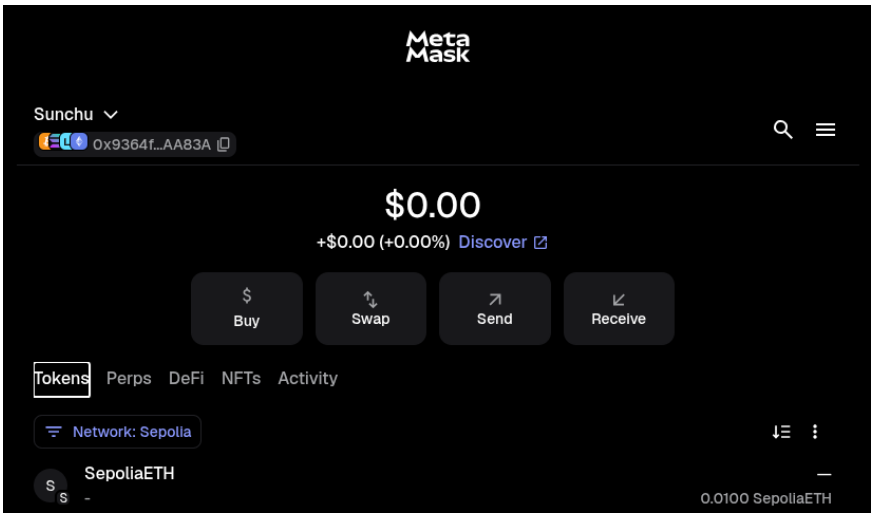


Fig. 7 – Receiver account showing 0.0100 SepoliaETH

4.7. Task 7 – Verify the Transaction on Etherscan

The transaction was verified using the Sepolia Etherscan block explorer. The transaction was shown with a successful status and confirmed block information.

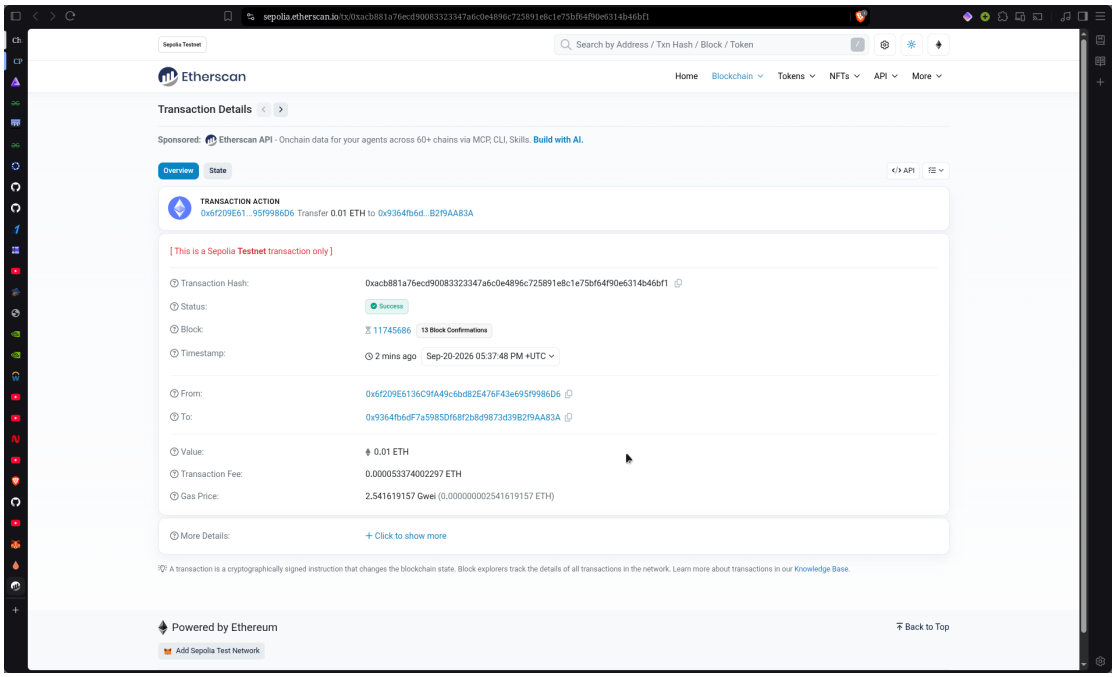


Fig. 8 – Transaction details on Sepolia Etherscan

The important transaction details recorded from the block explorer were:

Parameter	Observed Value
Network	Sepolia Testnet
Status	Success
Block	11745686
Confirmations	13
Value	0.01 ETH
Transaction Fee	0.000053374002297 ETH
Gas Price	2.541619157 Gwei

The transaction hash was:

```
0acb881a76ecd90083323347a6c0e4896c725891e8c1e75bf64f90e6314b46bf1
```

5. Transaction Flow

The complete flow of the experiment was:

```
Create MetaMask Account 1 (Nikhil)
             |
             v
Create MetaMask Account 2 (Sunchu)
             |
             v
Enable Sepolia Test Network
             |
             v
Obtain Test ETH from Faucet
             |
             v
Account 1 sends 0.01 SepoliaETH
             |
             v
Account 2 receives 0.0100 SepoliaETH
             |
             v
Verify Transaction on Sepolia Etherscan
```

6. Result

Two MetaMask accounts, `Nikhil` and `Sunchu`, were successfully created on the Sepolia test network. Test ETH was obtained through the Sepolia faucet, and **0.01 SepoliaETH** was successfully transferred from the Nikhil account to the Sunchu account. The transaction was confirmed on the Sepolia blockchain and independently verified using Etherscan.

7. Conclusion

The experiment demonstrated the creation and management of blockchain accounts using MetaMask and the transfer of cryptocurrency between two accounts. The use of the Sepolia test network allowed the transaction to be performed safely using test ETH. The successful confirmation on MetaMask and verification on Etherscan demonstrated how a cryptocurrency transaction is recorded and validated on a blockchain network.